

DETECTING MENTAL DISTRESS IN SOCIAL MEDIA POSTS USING MACHINE LEARNING

Final Report

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Higher Diploma in Science in Data Analytics

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Abstract

This project evaluates four traditional machine learning models, namely Logistic Regression, Naïve Bayes, Support Vector Machine (SVM), and Random Forest, for classifying psychological distress from social media text into five categories: Suicidal, Depressed, Anxious, Frustrated, and Others. Three TF-IDF feature representations were compared using Grid Search optimisation with 5-Fold Stratified Cross Validation. The results showed that combined n-gram representations generally outperformed unigram-only features. SVM achieved the highest accuracy of 0.8050 using UniBigram features, while Logistic Regression achieved the highest AUC-ROC score of 0.9524 using UniTrigram features. Considering the small performance difference between models, Logistic Regression with UniTrigram features was selected as the recommended model due to its competitive performance and greater interpretability. The findings demonstrate that traditional machine learning approaches remain effective for mental health text classification and highlight the importance of interpretable models in sensitive applications.

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1. Introduction

Mental health detection using machine learning has become an increasingly researched area due to the growing availability of user-generated online text. Social media platforms provide large volumes of written content where individuals may express emotions, experiences and indicators of psychological distress. Since the COVID-19 pandemic, concerns surrounding mental health have increased significantly, with the World Health Organization reporting a 25% rise in global anxiety and depression during the first year of the pandemic (WHO, 2022).

Natural Language Processing (NLP) and machine learning techniques provide methods for analysing unstructured text data and identifying patterns associated with psychological distress. This project investigates whether traditional machine learning text classification techniques can effectively classify mental distress categories from social media posts, with the aim of evaluating their suitability as a baseline approach for mental health text analysis.

1.1 Project Aims and Objectives

The aim of this project is to evaluate the effectiveness of traditional machine learning models in detecting mental distress from social media text.

The research question is:

"Can traditional machine learning models effectively classify psychological distress categories from social media text?"

The objectives of this study are:

- To preprocess and explore the MentalDistress dataset.
- To extract and compare TF-IDF feature representations using unigram, unigram-bigram and unigram-bigram-trigram configurations.
- To train and evaluate four classification models using Grid Search with 5-Fold Stratified Cross Validation.
- To compare interpretable and black-box model performance.
- To establish a robust traditional machine learning baseline for future transformer-based research.

1.2 Project Scope

This study focuses on English-language posts from the MentalDistress dataset (Prity et al., 2026), consisting of 10,100 annotated posts across five categories: Suicidal, Depressed, Anxious, Frustrated and Others. The research is limited to experimental evaluation of machine learning models, and real-world clinical deployment is outside the scope of this project.

2. Background and Literature Review

2.1 Background

Mental health text classification has become an important application of Natural Language Processing (NLP) due to the increasing availability of user-generated content on social media. Unlike structured clinical data, social media posts contain informal language, abbreviations, spelling variations and inconsistent grammar, making automated analysis more challenging. Consequently, effective classification requires appropriate preprocessing, feature

extraction and machine learning techniques to identify linguistic patterns associated with psychological distress. Traditional machine learning methods remain widely used because they are computationally efficient, interpretable and well suited to text classification tasks, making them appropriate baseline models for evaluating more advanced approaches.

2.2 Literature Review

The Cross-Industry Standard Process for Data Mining (CRISP-DM) provides a structured framework for developing machine learning projects through iterative stages of business understanding, data understanding, preparation, modelling, and evaluation (Chapman et al., 2000). Its flexible structure makes it suitable for NLP applications, where preprocessing and model optimisation often require repeated refinement.

Text preprocessing is a critical stage in NLP because raw text contains noise and linguistic variation that can affect classification performance. Techniques such as lowercasing, tokenisation, stop-word removal, and lemmatisation are commonly applied to improve text quality (Kallstenius et al., 2025). However, preprocessing in mental health applications requires careful consideration because linguistic patterns may contain meaningful psychological information.

Feature representation is another important factor in text classification. TF-IDF converts text into numerical representations by assigning higher importance to distinctive terms while reducing the influence of frequently occurring words (Sparck Jones, 1972; Manning et al., 2008). The use of n-grams further enhances representation by capturing contextual relationships between words, improving the ability of models to identify meaningful language patterns (Hameed et al., 2025).

Traditional supervised learning algorithms, including Logistic Regression, Support Vector Machine (SVM), Random Forest, and Naïve Bayes, remain effective approaches for mental health text classification, particularly when applied to medium-sized datasets with sparse text representations Hasan et al. (2026, online first). These models provide computationally efficient and interpretable baselines for comparison with more advanced approaches.

Model evaluation in multi-class classification requires multiple performance measures. While accuracy provides an overall measure of correctness, precision, recall, and F1-score provide a more balanced assessment of classification performance. Macro-averaged AUC-ROC further evaluates the model's ability to distinguish between different classes.

As machine learning is increasingly applied to sensitive healthcare domains, interpretability has become an important consideration. Explainable Artificial Intelligence (XAI) methods such as SHAP estimate feature contributions to model predictions, improving transparency and supporting trust in automated decision-making (Lundberg & Lee, 2017).

Recent studies have explored transformer-based models such as BERT and MentalBERT, which provide stronger contextual language understanding than traditional approaches (Ji et al., 2022). However, these models typically require greater computational resources and may provide less direct interpretability. Therefore, traditional machine learning approaches remain valuable as efficient and explainable baselines for mental health text classification research.

3. Methodology

This project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework across six phases, with deployment excluded as the study is purely analytical (Chapman et al., 2000). CRISP-DM was selected due to its iterative structure and direct applicability to text classification workflows, allowing findings from each phase to inform subsequent stages.

Phase	Description
Business Understanding	Define the problem, objectives and research question
Data Understanding	Explore dataset: class distribution, text length and word frequency
Data Preparation	NLP preprocessing, TF-IDF vectorisation and train test split
Modelling	Train and tune four models across three feature representations
Evaluation	Assess performance using test accuracy, precision, recall, F1 and AUC-ROC
Deployment	Excluded: purely analytical study

Model performance was evaluated using accuracy, precision, recall, F1-score, and macro-averaged AUC-ROC. Accuracy provides an overall measure of classification performance, while F1-score and AUC-ROC provide additional insight into model effectiveness across multiple classes. Five-fold cross-validation accuracy was also calculated as supporting evidence of model stability during evaluation.

4. CRISP-DM Implementation

4.1 Business Understanding

Mental health monitoring is an important research area due to the increasing need for scalable methods to identify psychological distress. Social media platforms provide large amounts of user-generated text that may contain indicators of emotional difficulties, creating opportunities for machine learning approaches to support mental health analysis. This project addresses the need to evaluate whether traditional machine learning models can effectively classify mental distress categories from social media posts while maintaining interpretability.

The key stakeholders include mental health professionals, healthcare organisations and researchers developing artificial intelligence solutions for mental health applications. For these stakeholders, predictive performance alone is insufficient; understanding the reasons behind model decisions is essential for responsible use in sensitive domains. Therefore, this project aims to establish an interpretable traditional machine learning baseline that can support future research and comparison with more advanced approaches.

4.2 Data Understanding and Exploration

MentalDistress dataset was obtained from Mendeley Data (Prity et al., 2026) and contains 10,099 annotated social media posts across five classification categories: Suicidal, Depressed, Anxious, Frustrated and Others. Initial exploration identified two records with missing text values and 48 duplicate entries. These records were removed during data cleaning, resulting in a final dataset of 10,050 posts used for subsequent model development and evaluation.

Class distribution analysis, presented in Figure 1, showed relatively balanced representation across the five categories, with sample counts ranging from 1,774 instances in the Others category to 2,170 instances in the Suicidal category. Although no severe class imbalance was observed, class weighting was later applied during model training to further account for differences between categories.

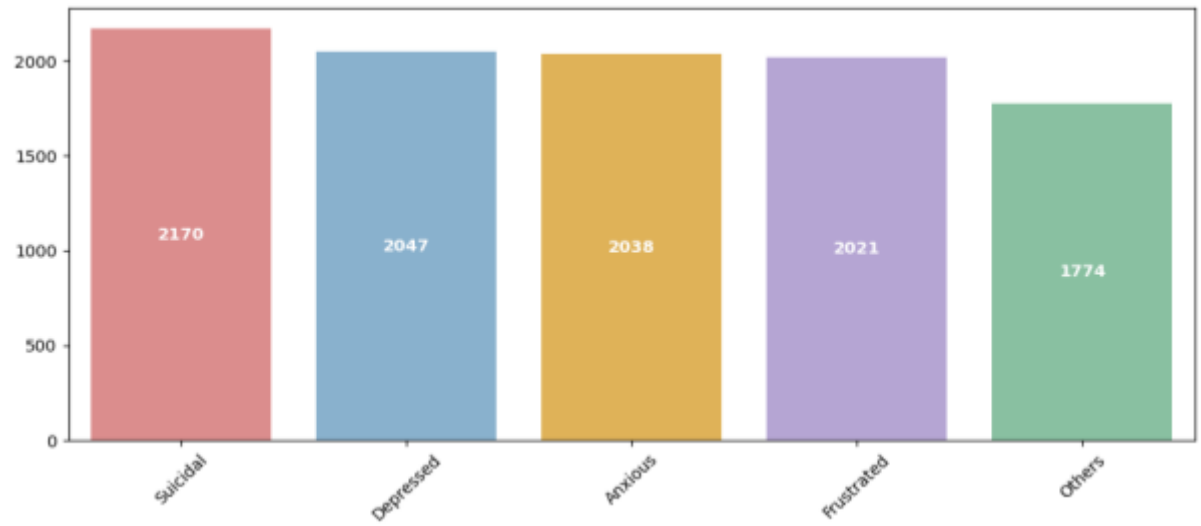


Figure 1 - Class Distribution

Text length analysis (Figure 2) showed right-skewed distributions across all categories, with most posts containing relatively short text and a smaller number of substantially longer posts. Anxious posts had the highest average length at 163 words, followed by Suicidal posts at 152 words and Depressed posts at 131 words. Frustrated and Others categories contained shorter average posts, with mean lengths of 75 and 52 words respectively. These variations highlight the diversity of social media text and the need for appropriate preprocessing before feature extraction.

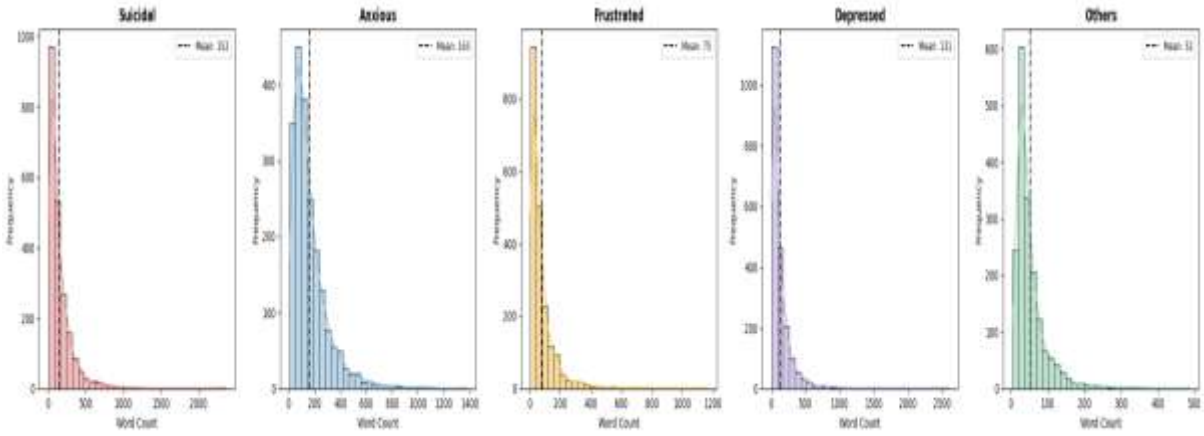


Figure 2 – Text Length Analysis Per Category

Frequency analysis was conducted on unigram, bigram and trigram features to identify common linguistic patterns within the dataset. Unigram features were dominated by frequent function words such as “i”, “to”, “and” and “the”, which provide limited contextual information when considered independently. In contrast, bigram and trigram analysis identified more meaningful multi-word expressions, including “do not”, “feel like”, “do not know” and “want to die”. These findings demonstrate the value of n-gram representations for capturing contextual relationships within social media text and support the comparison of unigram, unigram-bigram and unigram-bigram-trigram TF-IDF configurations.

4.3 Data Preparation and Feature Extraction

A standard NLP preprocessing pipeline was applied, including lowercasing, URL removal, contraction expansion, punctuation removal, stopword removal using NLTK and lemmatisation using the WordNet

Lemmatiser. As illustrated in Figure 3, word frequency analysis before preprocessing showed a dominance of common stopwords with limited discriminative value. Following preprocessing, more meaningful mental health-related terms and contextual phrases became more prominent, including patterns such as “health anxiety”, “want die” and “panic attack” within the bigram representation.

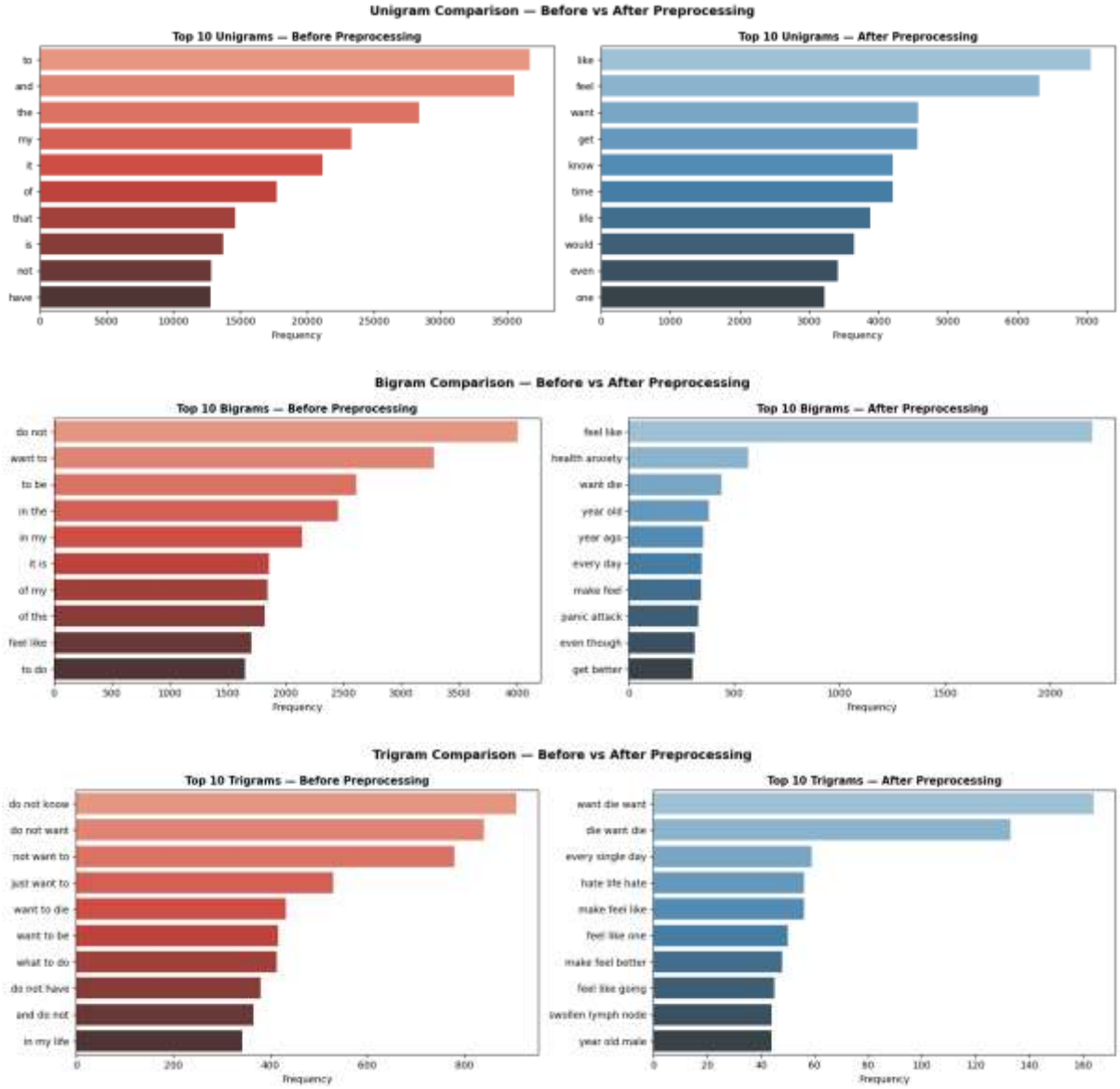


Fig 3 – Word Frequency before and after preprocessing

The dataset was first divided into training and testing subsets using an 80/20 stratified split, resulting in 8,040 training samples and 2,010 test samples. To prevent data leakage, all TF-IDF vectorisers were fitted exclusively on the training data and subsequently applied to the test set. TF-IDF representations were generated using a minimum document frequency of 3, a maximum document frequency threshold of 0.9, sublinear term frequency scaling (sublinear_tf=True), and smoothed inverse document frequency weighting (smooth_idf=True).

4.4 Model Development

Four classification models were developed and optimised: Logistic Regression, Support Vector Machine (SVM), Random Forest and Naïve Bayes. Each model was evaluated across the three TF-IDF feature representations using Grid Search with 5-Fold Stratified Cross Validation, resulting in twelve optimised model configurations.

Class weighting (class_weight='balanced') was applied to Logistic Regression, SVM and Random Forest to account for differences in class distribution. Naïve Bayes does not support class weighting and instead incorporates class probabilities through prior estimates. The complete Grid Search parameter configurations and cross-validation results are provided in Appendix A.

5. Results and Analysis

Tables 1, 2 and 3 present the performance of the twelve optimised model configurations evaluated using three TF-IDF feature representations: Unigram, UniBigram, and UniTrigram. Test set performance was evaluated using accuracy, precision, recall, F1-score, and macro-averaged AUC-ROC. In addition, 5-fold cross-validation accuracy was calculated on the training data to assess model stability across different data splits.

5.1 Unigram (1,1) Feature Results

Table 1 presents the performance of the four classification models using the unigram TF-IDF representation.

Model	Accuracy	Precision	Recall	F1 Score	CV Accuracy	AUC-ROC
Logistic Regression	0.7945	0.7948	0.7970	0.7935	0.7653	0.9515
Naive Bayes	0.7090	0.7296	0.7006	0.7010	0.6808	0.9324
SVM	0.8005	0.7995	0.8026	0.7971	0.7713	0.9433
Random Forest	0.7741	0.7795	0.7765	0.7648	0.7409	0.9411

Table 1 - Unigram (1,1) Test Set Results

The unigram representation provided a strong baseline across all evaluated models. SVM achieved the highest test accuracy (0.8005) and F1-score (0.7971), while Logistic Regression achieved the highest AUC-ROC score (0.9515). These results indicate that individual word features can effectively capture patterns associated with mental distress categories, although contextual information is limited.

The cross-validation results were consistent with the test performance, with SVM achieving the highest CV accuracy (0.7713), demonstrating stable performance across different training subsets.

5.2 UniBigram (1,2) Feature Results

Table 2 presents the classification performance when unigram and bigram features were combined.

Model	Accuracy	Precision	Recall	F1 Score	CV Accuracy	AUC-ROC
Logistic Regression	0.7955	0.7959	0.7979	0.7949	0.7703	0.9524
Naive Bayes	0.7214	0.7215	0.7165	0.7166	0.6981	0.9301
SVM	0.8050	0.8044	0.8068	0.8011	0.7715	0.9450
Random Forest	0.7791	0.7855	0.7819	0.7721	0.7310	0.9387

Table 2 - UniBigram (1,2) Test Set Results

The UniBigram representation improved performance across most models by introducing additional contextual information through word combinations. SVM achieved the strongest predictive performance, obtaining the highest accuracy (0.8050), F1-score (0.8011), and CV accuracy (0.7715).

Logistic Regression achieved the highest AUC-ROC score (0.9524), indicating strong class discrimination ability.

These results suggest that incorporating short word sequences improves classification performance compared with individual word features alone.

5.3 UniTrigram (1,3) Feature Results

Table 3 presents the classification performance using the combined unigram, bigram and trigram representation.

Model	Accuracy	Precision	Recall	F1 Score	CV Accuracy	AUC-ROC
Logistic Regression	0.7970	0.7979	0.7994	0.7966	0.7684	0.9524
Naive Bayes	0.7189	0.7202	0.7140	0.7140	0.6979	0.9296
SVM	0.8045	0.8039	0.8062	0.8004	0.7716	0.9449
Random Forest	0.7741	0.7783	0.7763	0.7655	0.7341	0.9379

Table 3 - UniTrigram (1,3) Test Set Results

The UniTrigram representation produced only marginal improvements compared with UniBigram features. Logistic Regression achieved its best performance using this representation, with an accuracy of 0.7970, F1-score of 0.7966, and AUC-ROC score of 0.9524. However, SVM performance remained similar to the UniBigram configuration.

The limited improvement suggests that increasing n-gram complexity introduced additional feature dimensionality and sparsity without providing substantial benefits for classification performance.

5.4 Comparison of Feature Representations and Models

Across all experiments, n-gram representations generally improved performance compared with unigram-only features, demonstrating the value of contextual information for mental distress classification. The largest improvement was observed when moving from Unigram to UniBigram features, where SVM accuracy increased from 0.8005 to 0.8050. This suggests that combining individual terms with short word sequences provides additional information for identifying patterns associated with different psychological categories.

UniBigram provided the most effective balance between contextual information and feature complexity. Although UniTrigram introduced longer word relationships, the improvement was limited compared with UniBigram. This indicates that increasing n-gram complexity may introduce additional feature sparsity and dimensionality without producing substantial performance gains.

Among the evaluated models, SVM achieved the strongest predictive performance, obtaining the highest accuracy (0.8050), F1-score (0.8011), and CV accuracy (0.7715) using UniBigram TF-IDF features. This demonstrates the effectiveness of SVM for handling high-dimensional sparse text representations. Logistic Regression achieved comparable performance and obtained the highest AUC-ROC scores, indicating strong class discrimination capability while providing greater interpretability through feature coefficients.

Naïve Bayes achieved lower performance across most evaluation metrics, likely due to its assumption that features are independent, limiting its ability to capture relationships between words. Random Forest also performed below the linear models, suggesting that tree-based approaches may be less suitable for sparse TF-IDF feature spaces.

Overall, SVM provided the best predictive performance, while Logistic Regression offered a stronger balance between performance, interpretability, and suitability for sensitive mental health text classification tasks.

5.5 Error Analysis

Error analysis was conducted using the confusion matrix of the recommended model to examine classification patterns between psychological categories. The results showed that most samples were correctly classified, while some misclassification occurred between categories with overlapping linguistic characteristics. This is expected because categories such as depression and suicidal ideation may share similar expressions related to emotional distress and hopelessness. The confusion matrix is provided in Appendix B.

5.6 Interpretability Analysis

Interpretability analysis was conducted to understand which linguistic features influenced model predictions. This is particularly important in mental health text classification, where understanding the reasoning behind predictions is valuable.

Logistic Regression coefficients were analysed because the model provides direct insight into feature importance. The results showed that different psychological categories were associated with distinct language patterns. For example, anxiety-related posts contained more clinical terms, while suicidal and depressive categories showed stronger associations with negative emotional expressions and indicators of hopelessness. The Others category was mainly associated with positive social language.

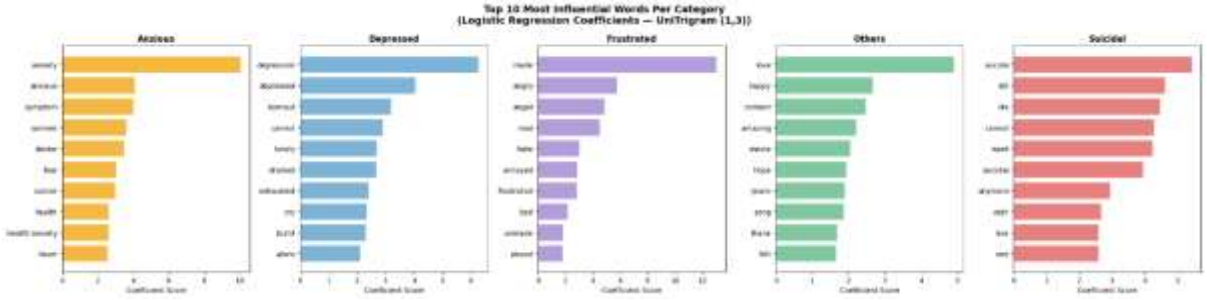


Figure 4 - Logistic Regression feature coefficients using UniTrigram TF-IDF features.

Although SVM achieved the highest predictive performance, it is less interpretable because feature contributions are not directly available. Therefore, SHAP analysis was applied to provide an explanation of SVM predictions. The SHAP results showed that SVM relied on similar vocabulary patterns identified by Logistic Regression, including terms related to anxiety, emotional distress, and social isolation.

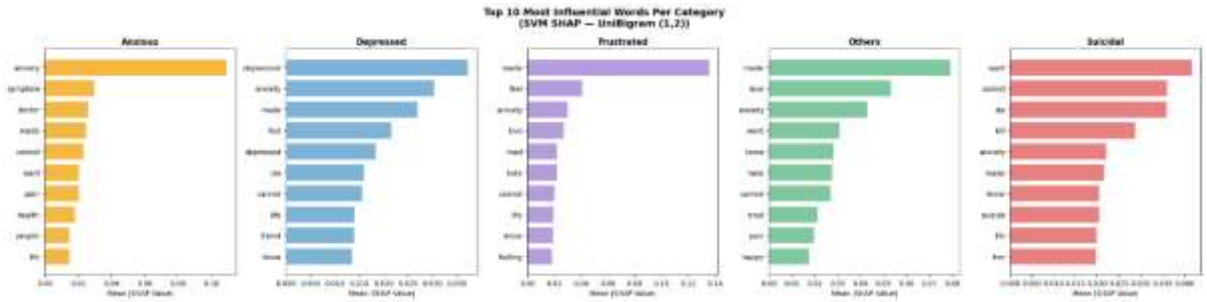


Figure 5 - SHAP feature importance for SVM using UniBigram TF-IDF features.

The consistency between Logistic Regression coefficients and SVM SHAP explanations suggests that both models learned meaningful psychological language patterns rather than relying on unrelated features. Additional interpretability results for other feature representations are provided in Appendix C.

5.7 Recommended Model

Based on the overall evaluation, Logistic Regression using the UniTrigram TF-IDF representation is recommended as the preferred model for this project. Although SVM achieved the highest accuracy

and F1-score, Logistic Regression achieved comparable performance while providing greater interpretability and transparency. In sensitive applications such as mental health classification, understanding the reasoning behind model predictions is important because interpretable models allow researchers and stakeholders to examine the factors influencing classification decisions (Rudin, 2019).

Logistic Regression with UniTrigram TF-IDF achieved a test accuracy of 0.7970, an F1-score of 0.7966, and an AUC-ROC score of 0.9524. These results demonstrate that the model can effectively classify mental distress categories while allowing feature contributions to be directly analysed through model coefficients. The small performance difference between Logistic Regression and SVM suggests that the additional complexity of SVM provides limited practical improvement for this dataset.

Therefore, Logistic Regression provides the best balance between predictive performance, interpretability, and suitability as a traditional machine learning baseline. This makes it an appropriate model for future comparisons between traditional machine learning approaches and more advanced transformer-based methods.

6. Discussion

6.1 Research Question Evaluation

This study investigated whether traditional machine learning models can effectively classify psychological distress categories from social media text. The results demonstrate that TF-IDF-based traditional approaches can achieve effective multi-class classification performance, with the best-performing models achieving accuracy close to 80%. These findings indicate that traditional machine learning methods remain useful as baseline approaches for mental health text classification.

The results also highlight the importance of contextual information in social media text classification. UniBigram representations generally achieved stronger performance than unigram features, suggesting that combinations of words provide additional information for identifying patterns associated with different distress categories. However, the limited improvement from UniTrigram features indicates that increasing feature complexity does not always lead to better performance due to increased feature sparsity.

SVM achieved the strongest predictive performance, obtaining the highest accuracy and F1-score. However, Logistic Regression was selected as the recommended model because it achieved comparable performance while providing greater interpretability through feature coefficient analysis. This is particularly valuable in mental health applications, where understanding the reasoning behind model predictions is important. The findings demonstrate that traditional machine learning models can provide effective and interpretable solutions while serving as strong baselines for comparison with more advanced approaches.

6.2 Limitations and Future Work

This study has several limitations. The evaluation was conducted using a single English-language dataset, which may limit the generalisability of the findings to different populations, platforms, or languages. Additionally, social media text contains informal expressions, spelling variations, and limited contextual information, which can create challenges for accurate classification.

This study focused on traditional machine learning approaches and did not compare performance with transformer-based models. Future research could evaluate transformer-based approaches, such as BERT-based models, against the established baseline to determine whether increased model complexity provides meaningful performance improvements. Further work could also investigate larger and more diverse datasets, as well as additional explainability techniques to improve understanding of model decisions.

Overall, the findings demonstrate that traditional machine learning models can provide an effective, efficient, and interpretable baseline for mental distress classification from social media text.

7. Conclusion

This project evaluated whether traditional machine learning models can classify psychological distress categories from social media text. Using the MentalDistress dataset, three TF-IDF feature representations and four classification algorithms were compared using Grid Search optimisation with 5-Fold Stratified Cross Validation.

The results show that traditional machine learning approaches can effectively identify mental distress patterns. N-gram representations improved performance by capturing contextual information, with UniBigram providing the best balance between performance and feature complexity. Although SVM achieved the highest predictive performance, Logistic Regression was selected as the recommended model due to its competitive results and greater interpretability.

The findings demonstrate that traditional machine learning models remain valuable for mental health text classification, particularly when transparency and explainability are important. This study provides a strong baseline for future comparison with advanced transformer-based models. Further research should explore larger, more diverse datasets and improved explainability techniques.

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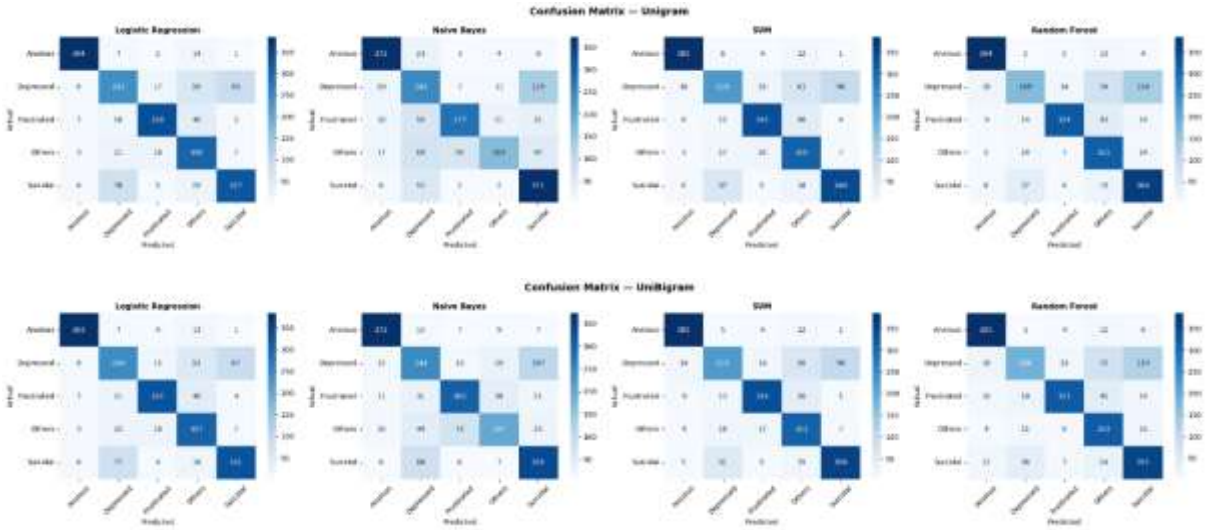
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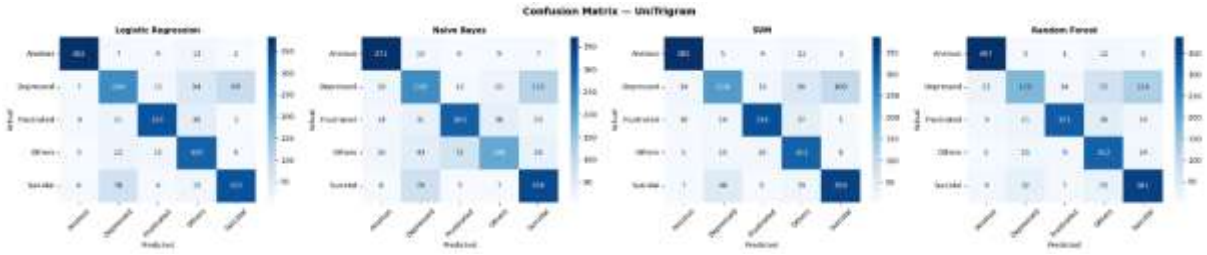
9. Appendices

Appendix A: Grid Search Hyperparameter Optimisation Results

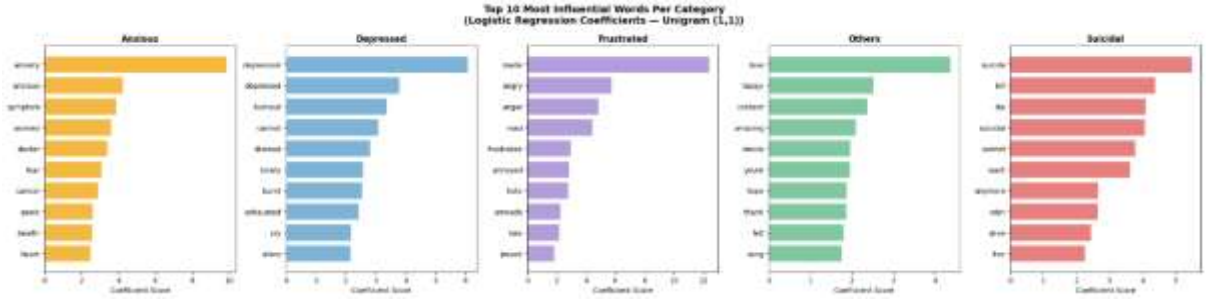
Feature	Model	Best Parameters	CV Accuracy
Unigram	Logistic Regression	C=1, solver=lbfgs	0.7653
Unigram	Naive Bayes	alpha=0.5	0.6808
Unigram	SVM	C=0.1	0.7713
Unigram	Random Forest	n_estimators=200, max_depth=None	0.7409
UniBigram	Logistic Regression	C=1, solver=lbfgs	0.7703
UniBigram	Naive Bayes	alpha=0.5	0.6981
UniBigram	SVM	C=0.1	0.7715
UniBigram	Random Forest	n_estimators=200, max_depth=None	0.7310
UniTrigram	Logistic Regression	C=1, solver=lbfgs	0.7684
UniTrigram	Naive Bayes	alpha=0.5	0.6979
UniTrigram	SVM	C=0.1	0.7716
UniTrigram	Random Forest	n_estimators=200, max_depth=None	0.7341

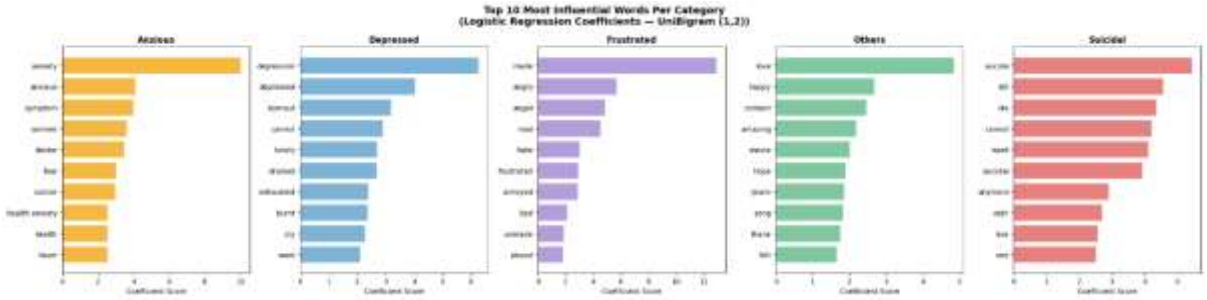
Appendix B: Confusion Matrix Analysis

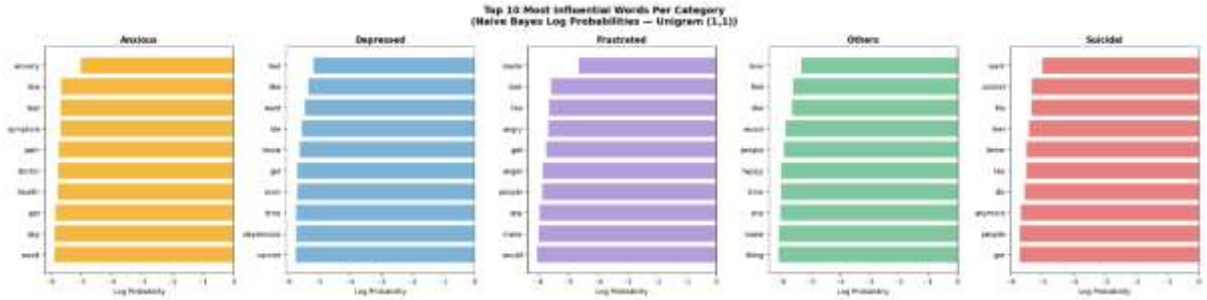


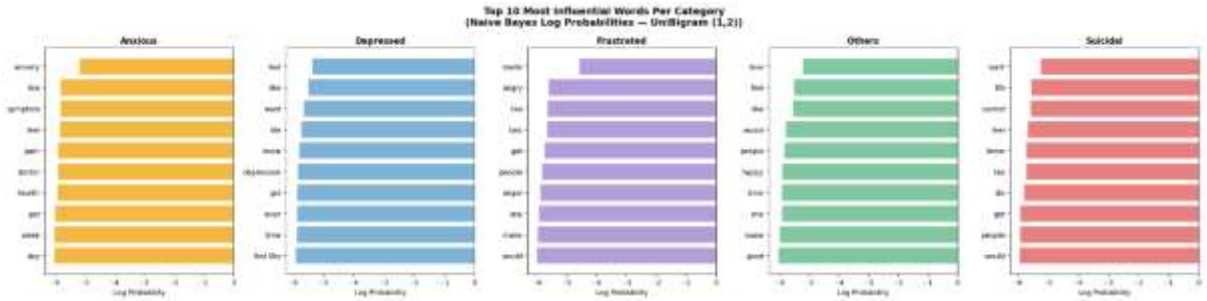


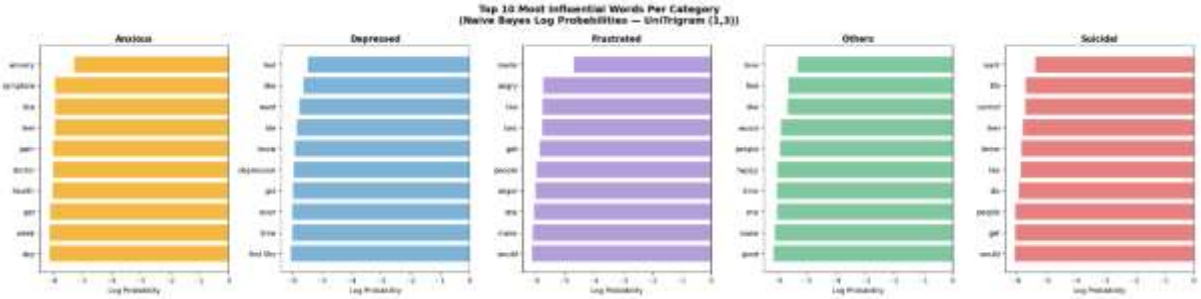
Appendix C: Additional Interpretability Results

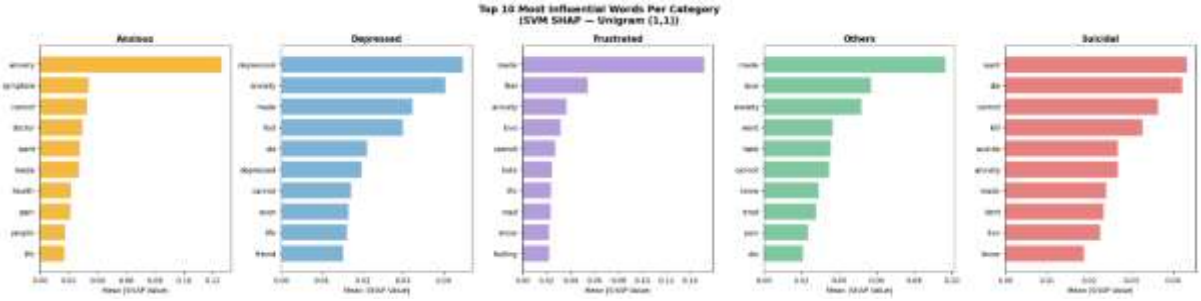


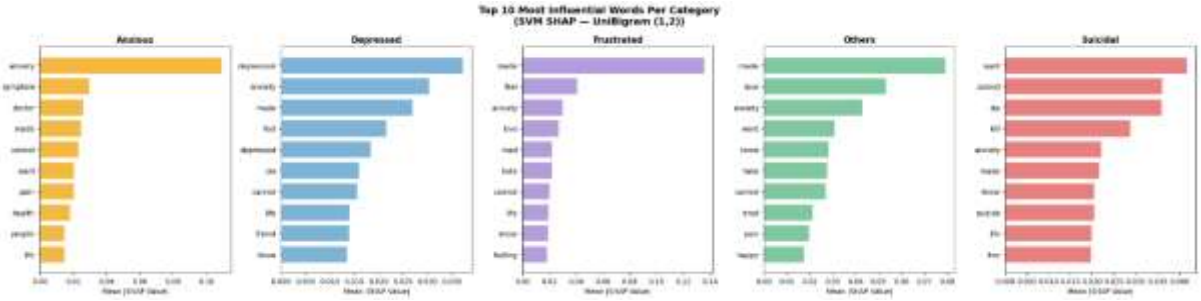







Top 10 Most Influential Words Per Category
(Naive Bayes Log Probabilities — UniTrigram (1,3))
Anxious
Depressed
Frustrated
Others
Suicidal


Top 10 Most Influential Words Per Category
(SVM SHAP — Unigram (1,1))
Anxious
Depressed
Frustrated
Others
Suicidal


Top 10 Most Influential Words Per Category
(SVM SHAP — UniBigram (1,2))
Anxious
Depressed
Frustrated
Others
Suicidal